

[Tutorials ▼](#)[References ▼](#)[Exercises ▼](#)[Certificates ▼](#)[In](#)[HTML](#) [CSS](#) [JAVASCRIPT](#) [SQL](#) [PYTHON](#) [JAVA](#) [PHP](#) [HOW TO](#) [W3.CSS](#) [C](#) [C](#) [>](#)

NumPy Tutorial

[NumPy HOME](#)[NumPy Intro](#)[NumPy Getting Started](#)[NumPy Creating Arrays](#)[NumPy Array Indexing](#)[NumPy Array Slicing](#)[NumPy Data Types](#)[NumPy Copy vs View](#)[NumPy Array Shape](#)[NumPy Array Reshape](#)[NumPy Array Iterating](#)[NumPy Array Join](#)[NumPy Array Split](#)[NumPy Array Search](#)[NumPy Array Sort](#)[NumPy Array Filter](#)

NumPy Random

[Random Intro](#)[Data Distribution](#)[Random Permutation](#)[Seaborn Module](#)[Normal Distribution](#)[Binomial Distribution](#)[Poisson Distribution](#)[Uniform Distribution](#)[Logistic Distribution](#)[Multinomial Distribution](#)[Exponential Distribution](#)[Chi Square Distribution](#)[Rayleigh Distribution](#)[Pareto Distribution](#)

ADVERTISEMENT

NumPy Array Slicing

[< Previous](#)[Next >](#)

Slicing arrays

Slicing in python means taking elements from one given index to another given index.

We pass slice instead of index like this:

`[start:end]` .

We can also define the step, like this:

`[start:end:step]` .

If we don't pass start its considered 0

If we don't pass end its considered length of array in that dimension

If we don't pass step its considered 1

Example

[Get your own Python Server](#)

Slice elements from index 1 to index 5 from the following array:

```
import numpy as np
```

```
arr = np.array([1, 2, 3, 4, 5, 6, 7])  
  
print(arr[1:5])
```

Try it Yourself »

Note: The result *includes* the start index, but *excludes* the end index.

Example

Slice elements from index 4 to the end of the array:

```
import numpy as np  
  
arr = np.array([1, 2, 3, 4, 5, 6, 7])  
  
print(arr[4:])
```

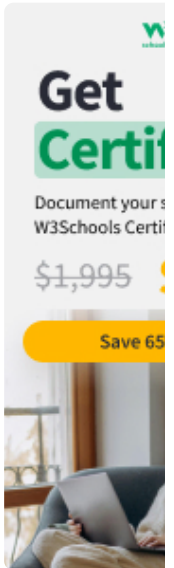
Try it Yourself »

Example

Slice elements from the beginning to index 4 (not included):

```
import numpy as np  
  
arr = np.array([1, 2, 3, 4, 5, 6, 7])  
  
print(arr[:4])
```

Try it Yourself »



COLOR
PICKER



ADVERTISEMENT

ADVERTISEMENT

[REMOVE ADS](#)

Negative Slicing

Use the minus operator to refer to an index from the end:

Example

Slice from the index 3 from the end to index 1 from the end:

```
import numpy as np  
  
arr = np.array([1, 2, 3, 4, 5, 6, 7])  
  
print(arr[-3:-1])
```

[Try it Yourself »](#)

[REMOVE ADS](#)

STEP

Use the **step** value to determine the step of the slicing:

Example

Return every other element from index 1 to index 5:

```
import numpy as np
```

```
arr = np.array([1, 2, 3, 4, 5, 6, 7])  
  
print(arr[1:5:2])
```

Try it Yourself »

Example

Return every other element from the entire array:

```
import numpy as np  
  
arr = np.array([1, 2, 3, 4, 5, 6, 7])  
  
print(arr[::2])
```

Try it Yourself »

Slicing 2-D Arrays

Example

From the second element, slice elements from index 1 to index 4 (not included):

```
import numpy as np  
  
arr = np.array([[1, 2, 3, 4, 5], [6, 7,  
8, 9, 10]])  
  
print(arr[1, 1:4])
```

Try it Yourself »

Note: Remember that *second element* has index 1.

Example

From both elements, return index 2:

```
import numpy as np

arr = np.array([[1, 2, 3, 4, 5], [6, 7, 8, 9, 10]])

print(arr[0:2, 2])
```

Try it Yourself »

Example

From both elements, slice index 1 to index 4 (not included), this will return a 2-D array:

```
import numpy as np

arr = np.array([[1, 2, 3, 4, 5], [6, 7, 8, 9, 10]])

print(arr[0:2, 1:4])
```

Try it Yourself »

Exercise [?]

Consider the following code: